Cross-Currency Dynamics in Cryptocurrencies

Abstract

When Silicon Valley Bank collapsed in March 2023, USDC traded as low as \$0.87, fragmenting Bitcoin prices across exchanges and quote currencies. Using 1-minute data from Binance, Coinbase, and Kraken, we decompose this fragmentation into a large stablecoin-driven component (USDC crisis-mean dispersion $D_t \approx +320$ bps) and a smaller adjusted residual ($B_t \approx +5$ bps, half-lives 0.54–1.10 minutes). A two-layer dynamic emerges: the adjusted basis mean-reverts in ≈ 0.6 minutes during crisis, while the USDC peg deviation half-life extends to ≈ 572 minutes—a $\sim 940\times$ gap localizing persistent dislocation to the peg layer, not cross-market arbitrage. Chow tests reject parameter stability at SVB onset (F=2,089 and 71, $p<10^{-16}$); USDC crisis excess kurtosis reaches 12.9 ($6\times$ P99 tail expansion); USDT trades at a $+58$ bps safe-haven premium; and cross-stablecoin correlation drops from 0.44 to 0.16—consistent with divergent reserve-credibility dynamics. Fiat-quoted BTC leads price discovery (Hasbrouck IS 0.73), while post-friction arbitrage opportunities narrow sharply. These findings map directly to the GENIUS Act's reserve-composition, timely-redemption, transparency, and bankruptcy provisions.

1 Introduction

Stablecoins are the primary quote currency on most crypto exchanges. Because they are not identical to bank deposits—redeemability, custody, and transparency differ—their prices can deviate from \$1, introducing a *stablecoin funding risk premium*: BTC trades at different nominal prices across quote currencies purely because of stablecoin-specific risk. Prior work on cross-exchange price dispersion has focused on fiat-quoted arbitrage and market efficiency, while studies of stablecoin de-pegs have largely examined daily price and market-cap dynamics. Less understood is how a sudden loss of reserve access propagates through high-frequency cross-currency bases, liquidity, and price discovery—the microstructure channel through which peg risk becomes trading risk. In 2025, the U.S. enacted the GENIUS Act, establishing a federal framework for dollar-pegged stablecoins with stricter reserve backing and transparency [11].

We analyze March 1–21, 2023 (UTC), spanning the SVB failure. When SVB collapsed on March 10, $\approx$\$3.3B of Circle's USDC reserves became inaccessible, and USDC de-pegged to \$0.874 on Kraken [3]. Figure 1 shows USDC falling sharply below par while USDT trades at a premium—consistent with stablecoin substitution under stress. This episode provides a high-frequency stress test of cross-currency basis behavior, liquidity fragmentation, and price discovery.

We make three contributions. First, we decompose stablecoin-driven fragmentation into a peg-deviation component and an adjusted residual, revealing a $\sim 940\times$ half-life gap that localizes persistent dislocation to the reserve-access layer. Second, we document pronounced asymmetry between USDC and USDT—in tail risk, volatility, and cross-stablecoin correlation—that is inconsistent with a common-shock explanation. Third, we map these empirical findings to specific provisions of the GENIUS Act, providing a quantitative lens for evaluating stablecoin regulation.

The remainder of this paper is organized as follows. Section 2 describes the data and methodology. Section 3 presents the empirical results across five dimensions: dispersion decomposition, cross-exchange dispersion, market microstructure, price discovery, and arbitrage. Section 4 interprets the findings through a regulatory lens, and Section 5 concludes.

2 Data and Methodology

2.1 Data sources and preprocessing

We collect 1-minute OHLCV candles for BTC quoted in USD, USDT, and USDC from Binance (BTC/USDT, BTC/USDC; dominant USDT volume), Coinbase (BTC/USD, BTC/USDT, USDT/USD; primary U.S. fiat gateway), and Kraken (BTC/USD, BTC/USDT, BTC/USDC, USDT/USD, USDC/USD; broadest stablecoin-to-fiat coverage). Stablecoin-to-USD conversion uses venue-specific legs. All series are aligned to a 1-minute UTC grid (Table 1). Coverage exceeds 97% for most series, though pre-SVB Kraken BTC/USDC falls to 89.6% with 42% forward-fill share. Short gaps (≤ 5 min) are forward-filled; larger gaps remain

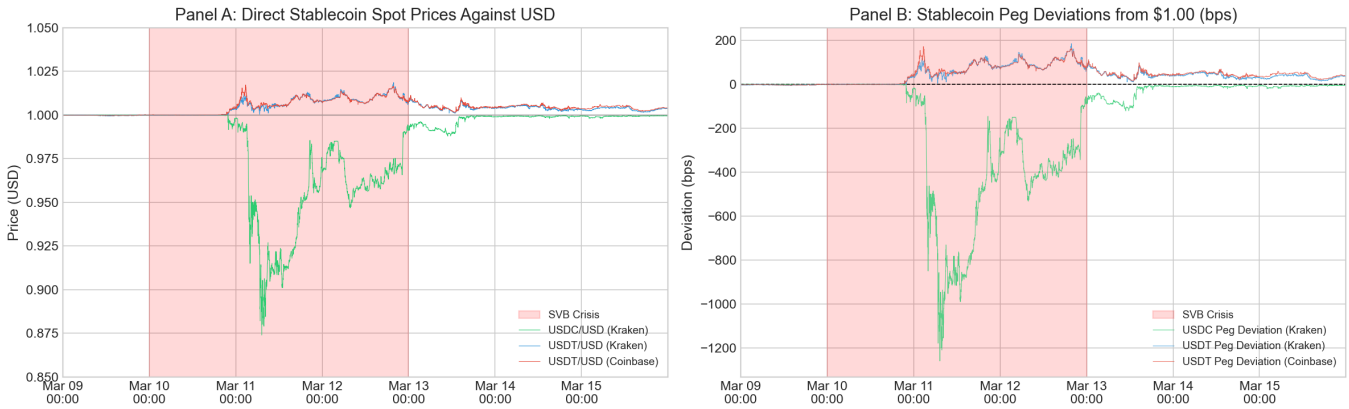

Figure 1: Stablecoin spot prices (A) and peg deviations in bps (B) during March 2023. USDC drops to \$0.874 while USDT trades at a premium. Shaded: SVB crisis window (March 10–13).

NaN. Core directional signs are stable under strict no-FF filtering (crisis USDC $D_t > 0$, USDT premium > 0), though some magnitudes shift on thinner samples. We interpret high-frequency levels as directional unless no-FF retention is adequate.

Table 1: Core series data coverage and forward-fill exposure. Coverage is the share of non-missing 1-minute observations on the unified UTC grid; forward-fill share is the percent of minutes filled by carry-forward (up to 5 minutes).

Pair	Coverage Overall (%)	Coverage Pre-SVB (%)	Coverage Crisis (%)	Coverage Post-SVB (%)	Forward-Fill Share Overall (%)
Kraken BTC/USD	99.98	100.00	100.00	99.95	0.18
Kraken BTC/USDT	98.44	97.36	99.81	99.06	30.84
Kraken BTC/USDC	94.05	89.62	99.79	96.57	42.07
Kraken USDC/USD	97.80	96.03	99.35	99.06	28.57
Kraken USDT/USD	99.98	100.00	100.00	99.95	0.91
Coinbase BTC/USD	99.10	97.90	100.00	100.00	0.06
Coinbase BTC/USDT	98.93	97.53	99.95	99.98	17.98
Coinbase USDT/USD	99.02	97.72	100.00	100.00	0.06
Binance BTC/USDT	100.00	100.00	100.00	100.00	0.00
Binance BTC/USDC	100.00	100.00	100.00	100.00	0.00

We proxy transaction frictions using the intra-candle range:

$$\text{Intra-minute Range (bps)} = \left(\frac{\text{High}_t - \text{Low}_t}{\text{Close}_t}\right) \times 10000. \tag{1}$$

This is a coarse range proxy that mechanically loads on volatility and microstructure noise; we interpret it cautiously as a friction indicator, not a quoted bid–ask spread.

2.2 Core quantitative objects

For stablecoin $Q \in \{\text{USDC}, \text{USDT}\}$, we track two objects. The *unadjusted dispersion*:

$$D_{Q,t} = \big(\ln(P_{\text{BTC}/Q,t}) - \ln(P_{\text{BTC/USD},t})\big) \times 10000, \tag{2}$$

and the *adjusted parity residual*:

$$B_{Q,t} = \big(\ln(P_{\text{BTC}/Q,t} \times P_{Q/\text{USD},t}) - \ln(P_{\text{BTC/USD},t})\big) \times 10000. \tag{3}$$

These satisfy $B_{Q,t} = D_{Q,t} + 10000 \cdot \ln(P_{Q/\text{USD},t})$: large de-pegs mechanically drive D_t while B_t captures the residual after marking to USD. Both are computed intra-exchange; cross-venue deviations are studied separately. Basis and dispersion are in log-bps; peg deviations, ranges, and volatility summaries are in arithmetic bps.

2.3 Mean-reversion dynamics

We model B_t as an OU process $dX_t = \theta(\mu - X_t)dt + \sigma dW_t$ and estimate the discrete AR(1) form $X_t = c + \rho X_{t-1} + \varepsilon_t$. Stationarity is checked with ADF tests (with large N, rejection is expected and serves as a baseline screen). The half-life uses the exact mapping:

$$\tau_{1/2} = \frac{\ln(2)\,\Delta t}{-\ln(\rho)}, \quad 0 < \rho < 1. \tag{4}$$

If $\rho \leq 0$ or $\rho \geq 1$, half-life is undefined (NaN). The OU framework assumes Gaussian innovations; crisis-period B_t exhibits substantial excess kurtosis (Section 3.3), so we use the AR(1) model solely for its half-life mapping, not as a full distributional model. We report robustness at 5-minute sampling and with forward-filled minutes removed.

2.4 Price discovery framework

For the two primary Kraken channels (BTC/USD vs. BTC/USDC and BTC/USD vs. BTC/USDT), we use synchronized 1-minute log-price levels, dropping forward-filled minutes. We test for cointegration with Johansen trace tests (constant restricted to the cointegration relation; lag order via BIC on the levels VAR) [8, 4]. When rank $= 1$, we estimate:

$$\Delta \mathbf{p}_t = \alpha\beta^\top \mathbf{p}_{t-1} + \sum_{i=1}^{k_\Delta} \Gamma_i \Delta \mathbf{p}_{t-i} + \mathbf{u}_t. \tag{5}$$

The market with smaller $|\alpha|$ adjusts less and is interpreted as the relative leader. We retain BH/FDR-corrected Granger tests on returns as secondary evidence only [6]; with 1-minute bars, asynchrony and large N can overstate economic lead–lag strength.

3 Empirical Results

3.1 Dispersion, adjusted residual, and mean reversion

Before SVB, both D_t and B_t hover near zero. During the crisis (March 10–13 UTC), USDC dispersion jumps to $+320$ bps (σ=323) while the adjusted residual remains small ($+5.3$ bps, σ=21). USDT moves oppositely: D_t=-57 bps, B_t=$+0.7$ bps (Table 2). Figure 2 visualizes this: D_t (orange) captures the peg shift while B_t (blue) remains tightly mean-reverting.

Two dynamics drive this pattern. USDC trades as low as \$0.874 on Kraken on March 11 ($\approx -1{,}260$ bps), mechanically pulling D_t upward; simultaneously, USDT trades at a premium averaging $+58.1$ bps on Kraken and $+59.7$ on Coinbase. Figure 3 visualizes this stablecoin substitution: crisis observations (red) migrate to the lower-right quadrant ($D_{\text{USDC}} > 0$, $D_{\text{USDT}} < 0$), and the unadjusted cross-stablecoin correlation flips from $+0.44$ to -0.42—consistent with traders rotating out of USDC into USDT under stress.

Table 2: Regime Statistics for Unadjusted Dispersion (D_t) and Adjusted Residual (B_t), Kraken

Regime	Series	Mean (bps)	Std (bps)	Mean Abs (bps)	N
Pre-SVB	USDC Kraken D_t (Unadjusted)	0.12	4.95	3.49	11615
Pre-SVB	USDC Kraken B_t (Adjusted)	-0.32	5.00	3.53	11197
Pre-SVB	USDT Kraken D_t (Unadjusted)	-0.54	4.49	3.17	12618
Pre-SVB	USDT Kraken B_t (Adjusted)	-0.60	4.33	3.02	12618
Crisis	USDC Kraken D_t (Unadjusted)	319.97	322.87	321.17	4311
Crisis	USDC Kraken B_t (Adjusted)	5.27	21.21	13.37	4283
Crisis	USDT Kraken D_t (Unadjusted)	-57.18	45.10	58.57	4312
Crisis	USDT Kraken B_t (Adjusted)	0.68	6.57	4.94	4312
Post-SVB	USDC Kraken D_t (Unadjusted)	11.22	21.95	14.27	12514
Post-SVB	USDC Kraken B_t (Adjusted)	0.45	9.06	6.68	12419
Post-SVB	USDT Kraken D_t (Unadjusted)	-30.26	13.29	30.28	12837
Post-SVB	USDT Kraken B_t (Adjusted)	-1.23	6.47	4.92	12837

Figure 2: D_t (orange) vs. B_t (blue) on Kraken during crisis. Top: USDC—D_t exceeds $+1{,}200$ bps while B_t stays within ± 150 bps. Bottom: USDT—D_t reflects the premium while $B_t \approx 0$. The gap is attributable to the stablecoin peg deviation.

HAC/Newey–West intervals (60-lag cap) confirm directional robustness [9]: USDC crisis D_t 95% CI $[245.7,\ 394.3]$ bps; crisis B_t CI $[3.6,\ 7.0]$ bps; USDT premiums positive on both Kraken ($[47.4,\ 68.8]$) and Coinbase ($[49.1,\ 70.3]$). Coinbase–Kraken BTC/USD remains significantly positive ($[1.7,\ 2.6]$ bps).

ADF rejects a unit root for all B_t channels ($p < 0.001$). Kraken USDC half-life: 0.98 min pre-SVB, 0.61 min crisis, 0.79 min post-SVB. Figure 4 shows robustness across 1m/5m sampling and all/no-FF filters: USDC crisis half-life ranges from 0.54 to 1.10 min; USDT crisis half-life is longer (0.92–3.06 min). Fast USDC mean reversion is robust across all specifications.

The OU estimation reveals a **two-layer dynamic** (Figure 5): B_t mean-reverts in ≈ 0.6 min during crisis ($\hat{\rho}=0.320$), while the USDC peg deviation persists with half-life ≈ 572 min ($\hat{\rho}=0.999$, ADF $p=0.44$); USDT peg deviations similarly persist (496–542 min). This $\sim 940\times$ gap shows cross-market correction is fast even in crisis, but the peg itself—driven by reserve-access risk—is where dislocation persists.

Figure 6 provides a granular minute-by-minute view of all channels during the peak crisis window. Panel A reveals that the USDC adjusted residual (green) exhibits extreme volatility—spikes exceeding $+150$ bps—while the USDT channels on both Kraken and Coinbase remain tightly bounded near zero throughout. This asymmetry reinforces the finding that the crisis is USDC-specific rather than a broad market dislocation. Panel B shows that cross-exchange differentials widen simultaneously, with the Coinbase–Kraken BTC/USD basis rising to $+60$ bps, suggesting stress propagation across venues—a pattern we examine in the next subsection.

3.2 Cross-exchange price dispersion

Having established that intra-exchange fragmentation is dominated by USDC peg risk, a natural question is whether this dislocation propagates *across* exchanges. We examine this by computing log-price differentials for BTC/USDT (Binance vs. Kraken; Coinbase vs. Kraken) and BTC/USD (Coinbase vs. Kraken). Caveat: USDT cross-exchange dispersion is in the USDT numeraire; a USD-based arbitrage requires an additional USDT/USD leg. Binance–Kraken BTC/USDT is near zero full-sample ($+0.007$ bps), consistent with tight integration. Coinbase–Kraken BTC/USD shows a persistent $+2.18$ bps premium, consistent with a structural U.S. fiat gateway effect.

During the crisis, cross-exchange differentials widen alongside intra-exchange deviations (Figure 7, Table 8), consistent with broad stress rather than a single-venue disruption. The Coinbase–Kraken fiat pre-

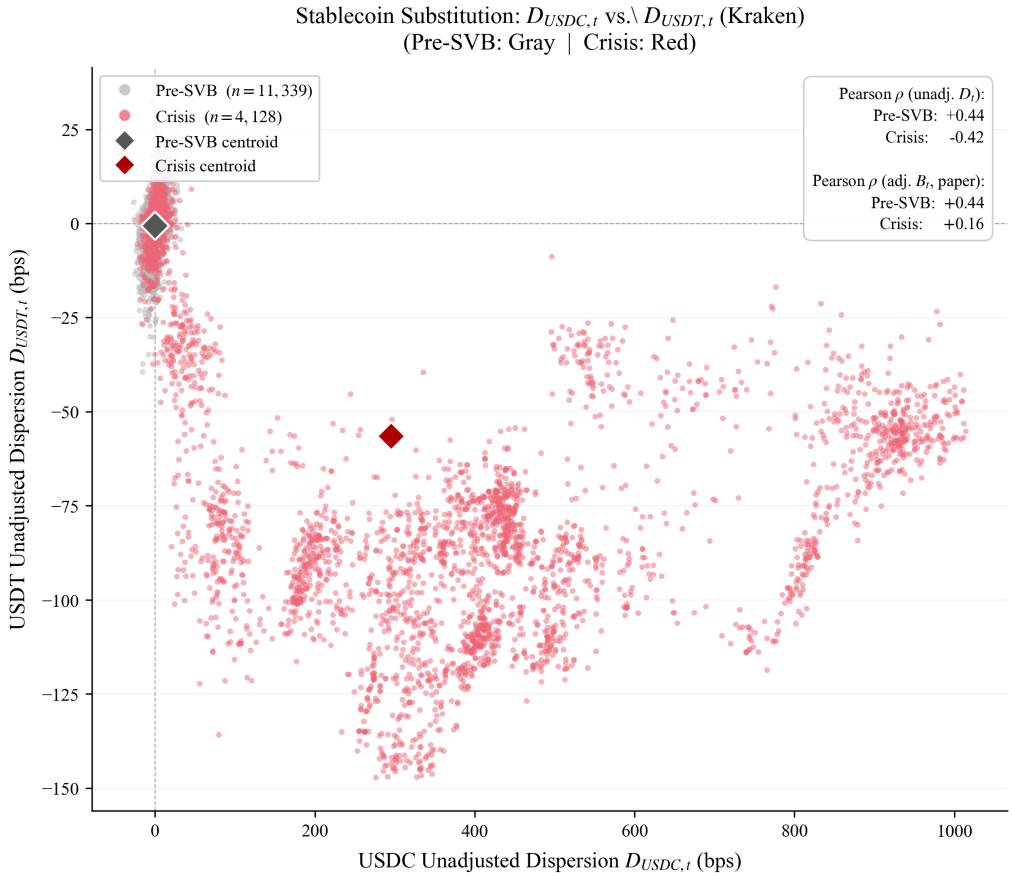

Figure 3: Stablecoin substitution: $D_{\text{USDC},t}$ vs. $D_{\text{USDT},t}$ on Kraken. Pre-SVB (gray) clusters near the origin; crisis (red) migrates to the lower-right quadrant. Unadjusted correlation flips from $+0.44$ to -0.42; adjusted B_t correlation drops from 0.44 to 0.16.

mium did not fully normalize post-crisis, suggesting it partly reflects structural microstructure differences that stress amplified.

3.3 Market microstructure dynamics

The cross-exchange widening documented above raises the question of whether liquidity conditions differ materially across channels. We quantify liquidity with the Roll (1984) effective spread [10] and Amihud (2002) ILLIQ ratio [1]. Table 3 and Figure 8 report regime means. Kraken BTC/USDC Roll spread jumps $23\times$ (1.10 to 25.64 bps), while BTC/USDT holds near 3 bps. Amihud confirms Coinbase BTC/USD is deepest and Kraken BTC/USDC least liquid. The two measures diverge in crisis: Roll widens while Amihud declines as volume surges.

Table 3: Roll (1984) effective spread (bps) and Amihud (2002) illiquidity ratio ($\times 10^{-6}$) by pair and regime. Roll spread estimated from daily serial covariance of 1-minute log returns; NaN days (non-negative covariance) excluded from means. N is the number of valid Roll days per regime; some regime means rest on few days and are indicative. $\text{ILLIQ}_t = |r_t|/\text{DollarVol}_t$, daily average.

Pair	Roll_{Pre}	$\text{Roll}_{\text{Crisis}}$	$\text{Roll}_{\text{Post}}$	N_{Pre}	N_{Crisis}	N_{Post}	$\text{ILLIQ}_{\text{Pre}}$	$\text{ILLIQ}_{\text{Crisis}}$	$\text{ILLIQ}_{\text{Post}}$
Kraken BTC/USD	1.260	—	—	4	0	0	0.232	0.050	0.068
Kraken BTC/USDT	3.390	2.850	2.670	1	1	1	8.299	7.266	9.074
Kraken BTC/USDC	1.100	25.640	2.070	2	2	2	31.970	13.282	23.530
Binance BTC/USDT	1.560	0.670	3.630	6	1	5	0.163	0.461	0.591
Coinbase BTC/USD	1.880	2.490	3.490	5	1	1	0.005	0.003	0.003

Daily dollar-volume depth proxies (Table 4) sharpen this picture: Coinbase BTC/USD is the deepest channel ($239M pre-SVB, rising to $664M post-SVB), while Kraken BTC/USDC is $26\times$ thinner in crisis ($15.7M vs. $407M). The depth gap constrains the scale at which arbitrageurs can trade the USDC dislocation, consistent with the limited post-friction profitability documented in Section 3.5.

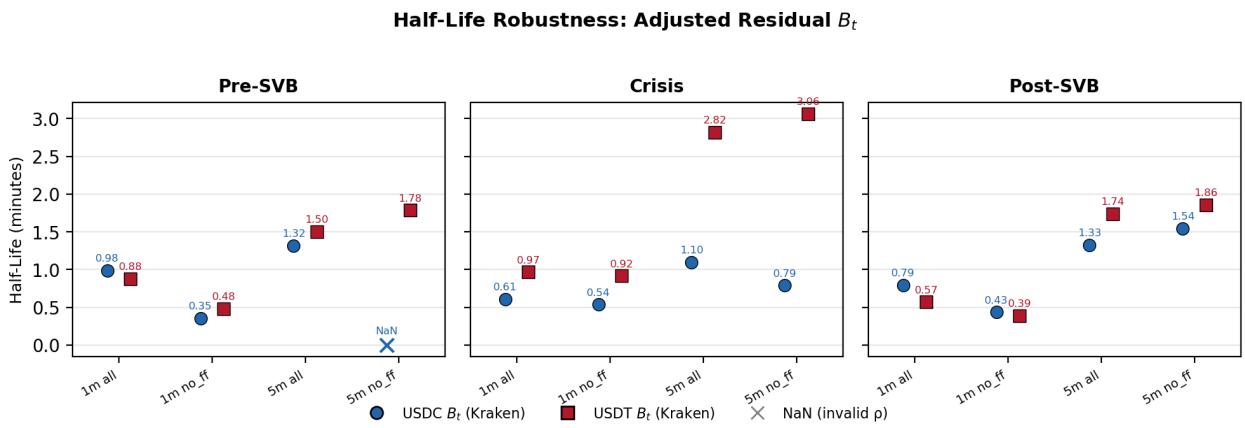

Figure 4: Half-life robustness for adjusted residual B_t: USDC (blue circles) vs. USDT (red squares) across four specifications (1m/5m $\times$ all/no-FF) and three regimes. USDC crisis half-life is consistently below 1.1 min; USDT crisis half-life reaches 3.1 min at 5m no-FF.

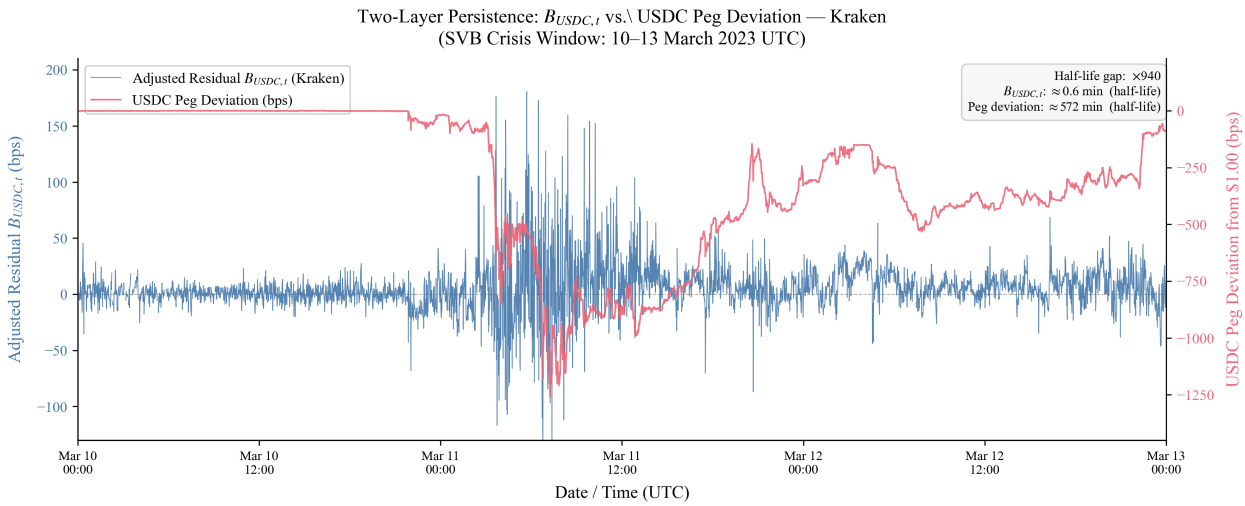

Figure 5: Two-layer persistence (March 10–13). $B_{\text{USDC},t}$ (blue, left) mean-reverts in $\approx$0.6 min; USDC peg deviation (red, right) persists for hours ($\approx$572 min). The $\sim$940$\times$ gap localizes dislocation to the peg layer.

Table 4: Daily traded dollar volume depth proxy by pair and regime (median, USD millions/day). Because historical L2 order-book snapshots are unavailable in the candle APIs used here, dollar volume and Amihud ILLIQ are used as complementary depth proxies.

Pair	Median DollarVol Pre-SVB (MM/day)	Median DollarVol Crisis (MM/day)	Median DollarVol Post-SVB (MM/day)
Kraken BTC/USD	59.50	160.36	203.77
Kraken BTC/USDT	5.55	13.84	17.37
Kraken BTC/USDC	2.56	15.65	5.52
Binance BTC/USDT	52.44	73.92	71.46
Coinbase BTC/USD	239.07	407.27	664.46

HAC regressions (Newey–West, 60 lags) of B_t on a crisis dummy, realized volatility, and range proxy (Table 5) show a significant USDC crisis coefficient ($+4.30$ bps, $p < 0.001$) [9]. Explanatory power is modest ($R^2 = 0.041$), so we interpret these as partial associations.

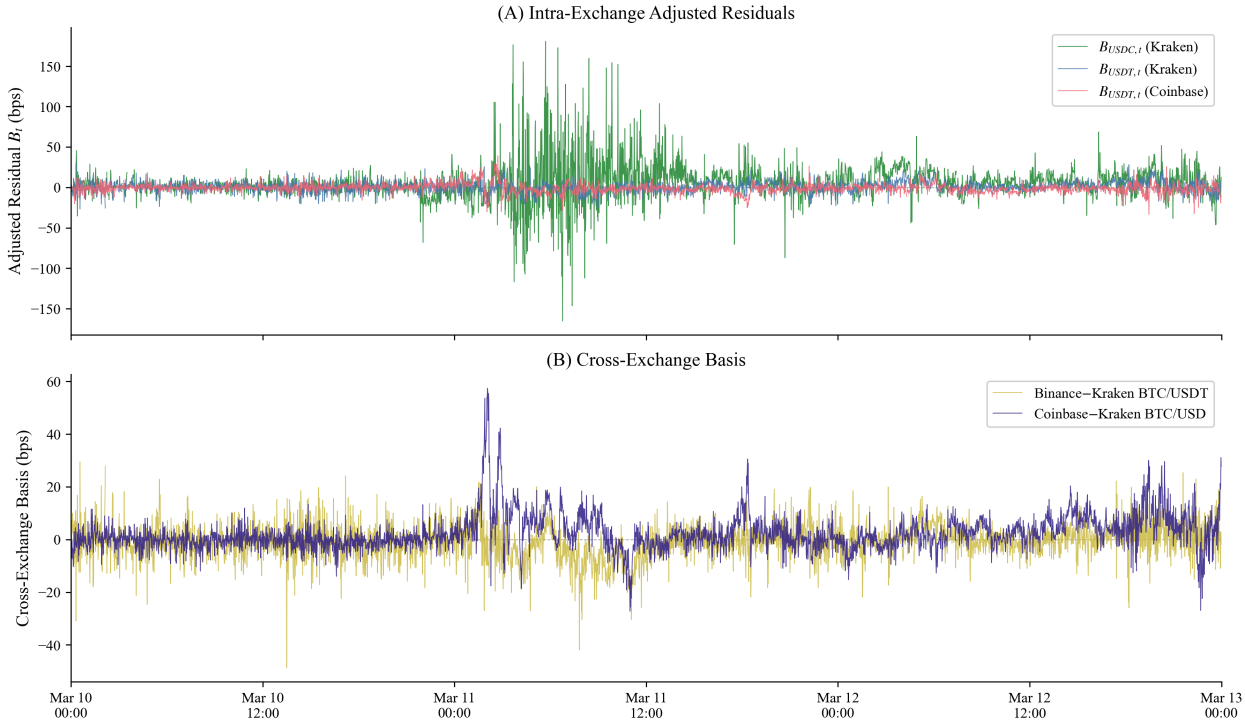

Figure 6: Minute-level crisis zoom (March 10–13 UTC). (A) Intra-exchange adjusted residuals: $B_{\text{USDC},t}$ (Kraken, green) spikes past ± 150 bps while USDT channels stay within ± 30 bps. (B) Cross-exchange basis: both Binance–Kraken BTC/USDT and Coinbase–Kraken BTC/USD widen, indicating stress propagation across venues.

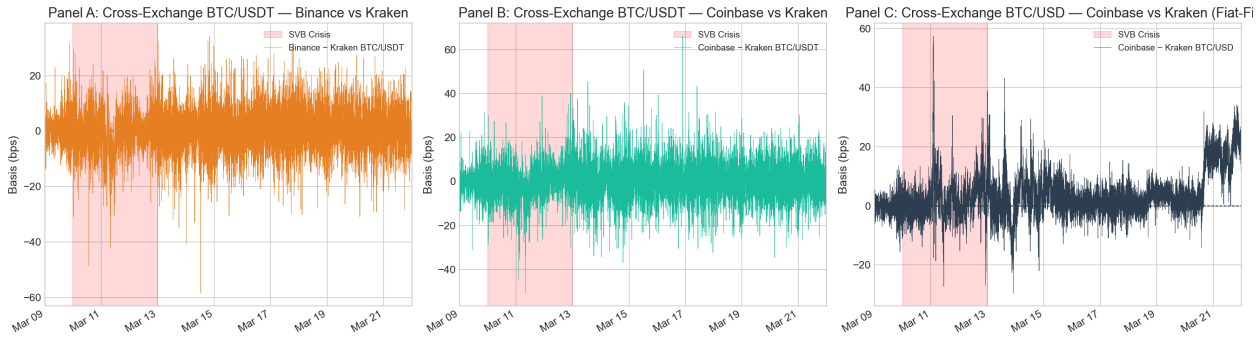

Figure 7: Cross-exchange basis (bps): BTC/USDT Binance–Kraken (A), Coinbase–Kraken (B), and BTC/USD Coinbase–Kraken (C). The fiat channel shows a persistent positive basis that remains elevated post-crisis, suggesting structural microstructure differences.

Table 5: HAC Regressions of Adjusted Residual B_t on Crisis Dummy, Realized Volatility, and Range Proxy (Kraken, Newey–West 60 lags)

	USDC Channel		USDT Channel	
	Coef.	p-value	Coef.	p-value
Constant	-0.079	0.833	-0.656	< 0.001
Crisis	$+4.295$	< 0.001	$+1.655$	< 0.001
RealizedVol (60m)	$+0.004$	0.928	-0.029	0.179
Range Proxy	$+0.048$	0.005	-0.013	0.082
R^2	0.041		0.012	
N	26,489		26,489	

OLS with HAC standard errors (Newey–West, 60 lags). Dependent variable: B_t (bps).

Figure 9 shows daily volume fragmentation: USDC share doubles during crisis (2.1% to 4.3%) as activity concentrates in Kraken BTC/USDC, while post-crisis USD dominance returns (86.9%).

Realized volatility (Figure 10) reveals pronounced asymmetry: Kraken BTC/USDC peaks at ≈ 650 bps/hr on March 11, roughly $2.1\times$ the crisis-average for BTC/USD and BTC/USDT—consistent with com-

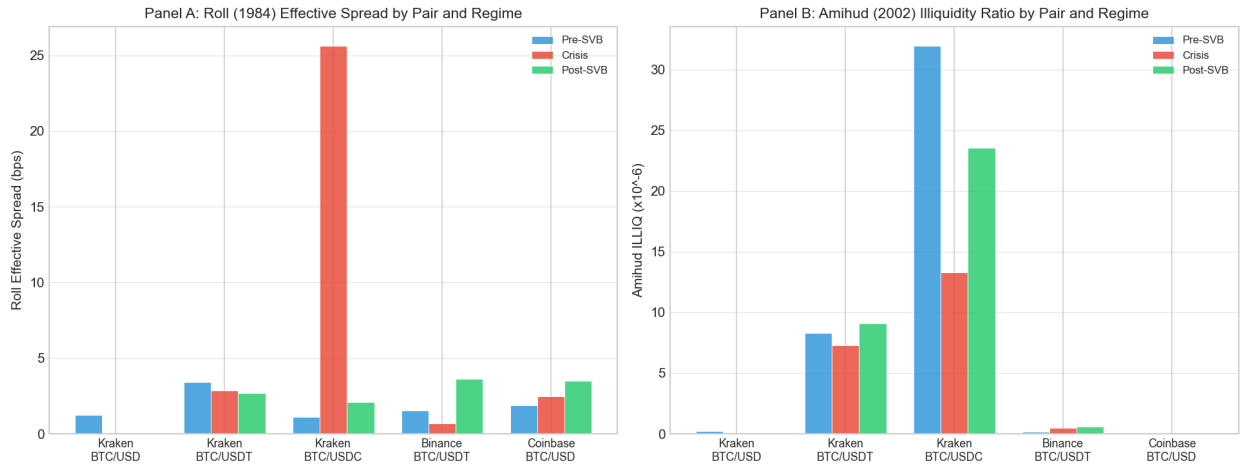

Figure 8: Roll (1984) effective spread (A) and Amihud (2002) ILLIQ ratio (B) by pair and regime. Kraken BTC/USDC shows a $23\times$ Roll spread blowout in crisis. Roll is undefined when serial covariance is non-negative; regime means for thin pairs rest on few valid days and are indicative.

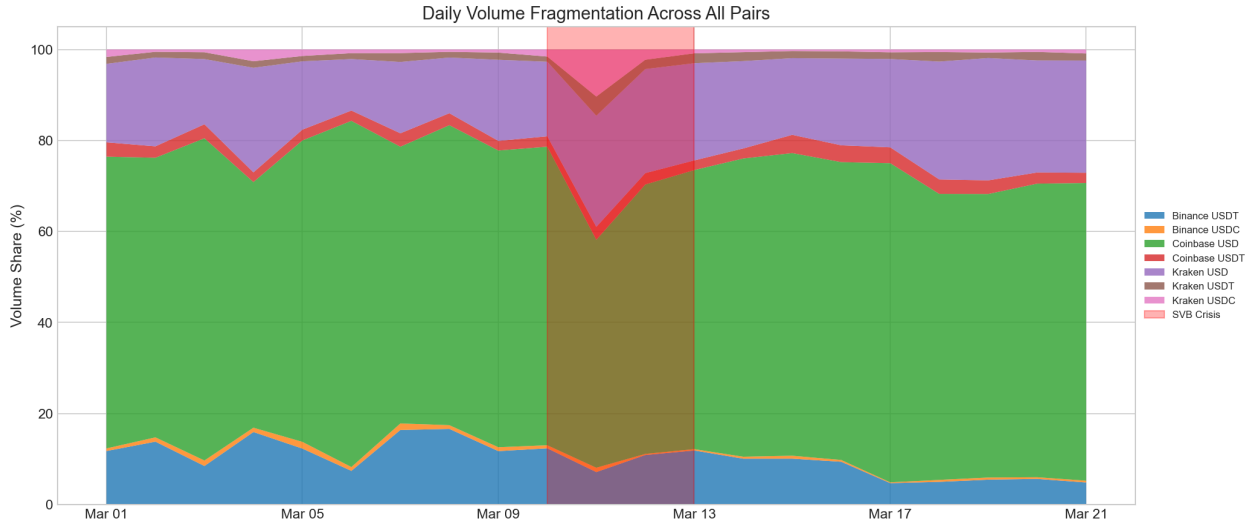

Figure 9: Daily volume fragmentation across all pairs. USDC share spikes during crisis as Kraken BTC/USDC activity surges; post-crisis, volume rotates back to USD dominance.

pounding asset risk and numeraire risk.

The distributional impact is stark (Figure 11): crisis USDC B_t develops tails beyond ± 100 bps versus ± 10 bps pre-SVB. For risk management, stablecoin positions carry latent peg-break tail risk that calm-calibrated VaR models materially understate.

Distributional robustness. Chow tests reject parameter stability at March 10 for D_t (F=2,089), B_t (F=230), and AR(1) dynamics (F=71; all $p<10^{-16}$), validating the regime split [2]. Table 6: USDC crisis B_t has excess kurtosis 12.9 and skewness $+1.01$ (vs. pre-SVB 3.8/$\approx$0); USDT crisis kurtosis is only 1.4—confirming USDC-specific tail risk. Jarque–Bera rejects normality throughout (USDC crisis JB=30,498). Cross-stablecoin correlation drops from 0.44 to 0.16 in crisis, consistent with divergent dynamics rather than a common shock.

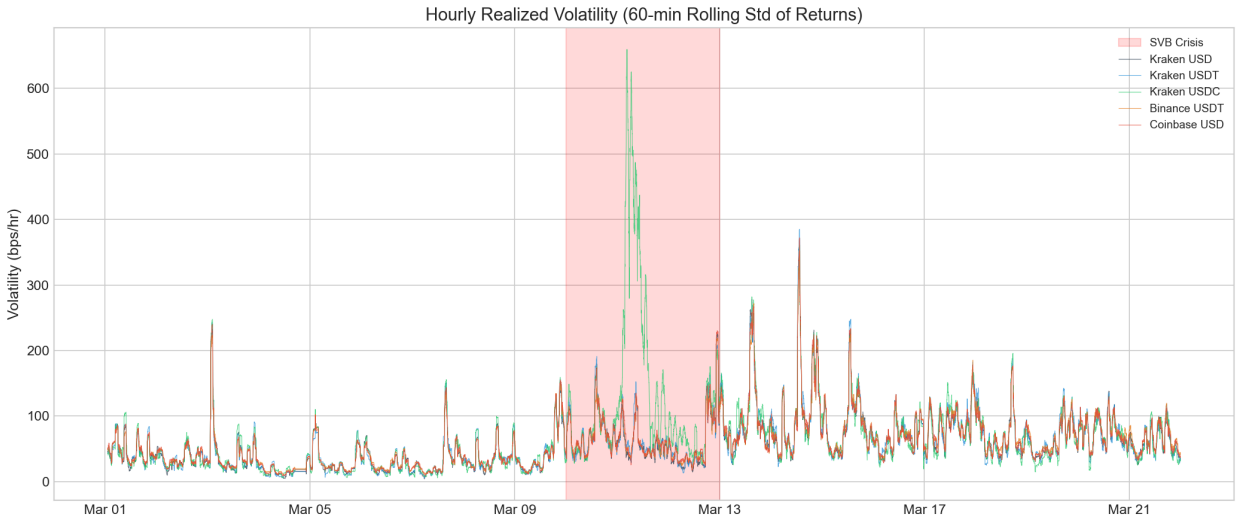

Figure 10: Hourly realized volatility (60-min rolling σ, $\times\sqrt{60}$) by pair. Kraken BTC/USDC (green) peaks at $\approx$650 bps/hr during crisis, roughly $2.1\times$ the USD and USDT channels, consistent with compounding asset and numeraire risk.

Table 6: Higher-moment diagnostics for adjusted residual B_t by regime. Excess kurtosis (Fisher) measures tail heaviness beyond Gaussian ($= 0$). Min/Max are in basis points.

Channel	Regime	Skewness	Ex. Kurt.	Min	Max
USDC	Pre-SVB	0.00	3.8	-36.9	32.0
USDC	Crisis	1.01	12.9	-165.2	180.6
USDC	Post-SVB	0.30	2.8	-43.2	79.4
USDT	Pre-SVB	-0.36	5.2	-36.4	34.5
USDT	Crisis	-0.11	1.4	-26.8	29.9
USDT	Post-SVB	-0.03	2.0	-37.8	36.2

Figure 12 visualizes the decorrelation across all channels: pre-SVB cross-channel correlations (B_{USDC}–B_{USDT}: 0.44) collapse in crisis (0.16) and partially recover post-SVB (0.38), while cross-exchange USDT channels remain tightly correlated throughout.

3.4 Price discovery

Given the liquidity asymmetry documented above—Coinbase BTC/USD is $26\times$ deeper than Kraken BTC/USDC—a natural question is where price discovery occurs across these fragmented channels.

Table 7: Johansen Cointegration and VECM Price Discovery (Primary Kraken Channels, No-FF Sample)

Channel	Rank	k_Δ	Trace$_{r=0}$	α_{USD}	α_{other}	IS$_{\text{USD}}$ [0.69, 0.77]	Leader
BTC/USD vs BTC/USDC	0	2	7.71	—	—	—	undetermined
BTC/USD vs BTC/USDT	1	8	27.03	0.0079	0.0118	0.73	BTC/USD

95% critical value for trace $r = 0$: 15.49. IS = Hasbrouck (1995) midpoint; bounds [0.69, 0.77].

Table 7 shows Kraken BTC/USD vs. BTC/USDT rejects no cointegration (rank $= 1$; stable at $k_\Delta \pm 1$). Hasbrouck (1995) IS bounds via both Cholesky orderings [7]: IS$_{\text{USD}} \in [0.69, 0.77]$, midpoint **0.73**—BTC/USD accounts for $\approx$73% of permanent price discovery. Bounds are tight (width 0.08), so leadership is not ordering-driven. Residual correlation 0.889 confirms high co-movement. For BTC/USD vs. BTC/USDC, no-FF does not reject rank $= 0$, consistent with long-run instability over this short window.

Figure 13 provides complementary visual evidence via orthogonalized impulse-response functions from a bivariate VAR on 1-minute returns (AIC-selected 10 lags; Cholesky ordering: BTC/USD first, consistent with its IS leadership; N=26,548 full-sample observations). A one-standard-deviation shock to BTC/USD transmits significantly to BTC/USDC (Panel C: $\approx$6 bps at impact, decaying within 3–4 minutes), while the reverse channel (Panel B) is statistically indistinguishable from zero—the 95% confidence band envelops the zero line at all horizons. This asymmetry is consistent with the Hasbrouck IS finding and confirms that fiat-quoted prices lead stablecoin-quoted prices in the information transmission chain.

BH/FDR-corrected Granger tests on returns provide secondary evidence: BTC/USD $\rightarrow$ BTC/USDC

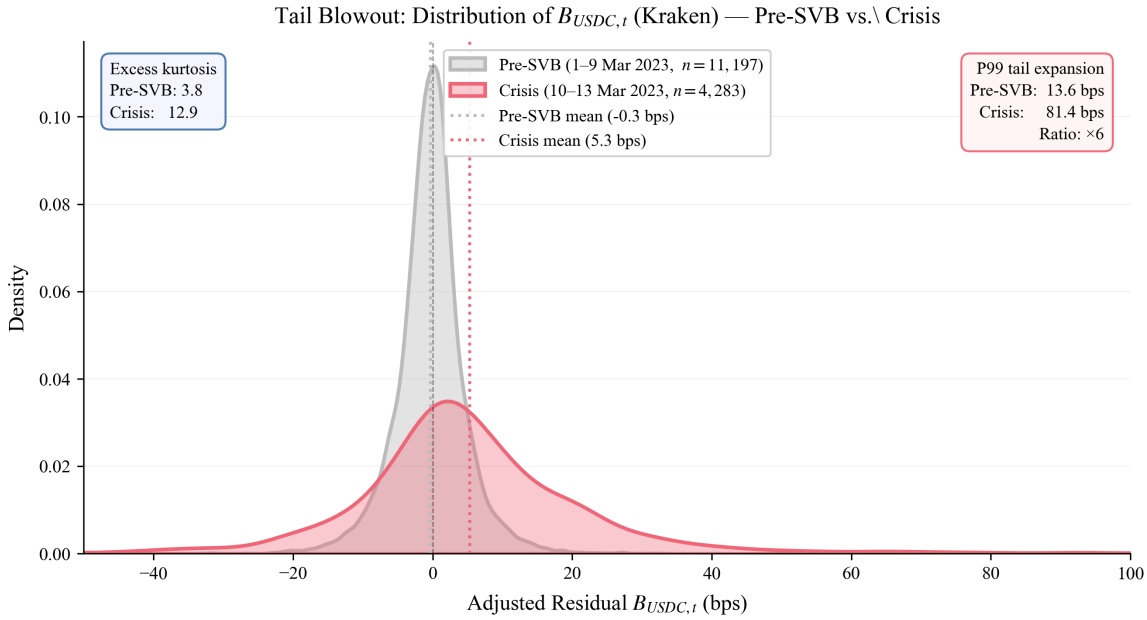

Figure 11: Distribution of the USDC adjusted residual $B_{\text{USDC},t}$ on Kraken: pre-SVB (gray) versus crisis (red). The crisis distribution exhibits a rightward shift, excess kurtosis of 12.9 (vs. 3.8 pre-SVB), and a $6\times$ expansion in the 99th percentile (from 13.6 to 81.4 bps), confirming USDC-specific tail risk.

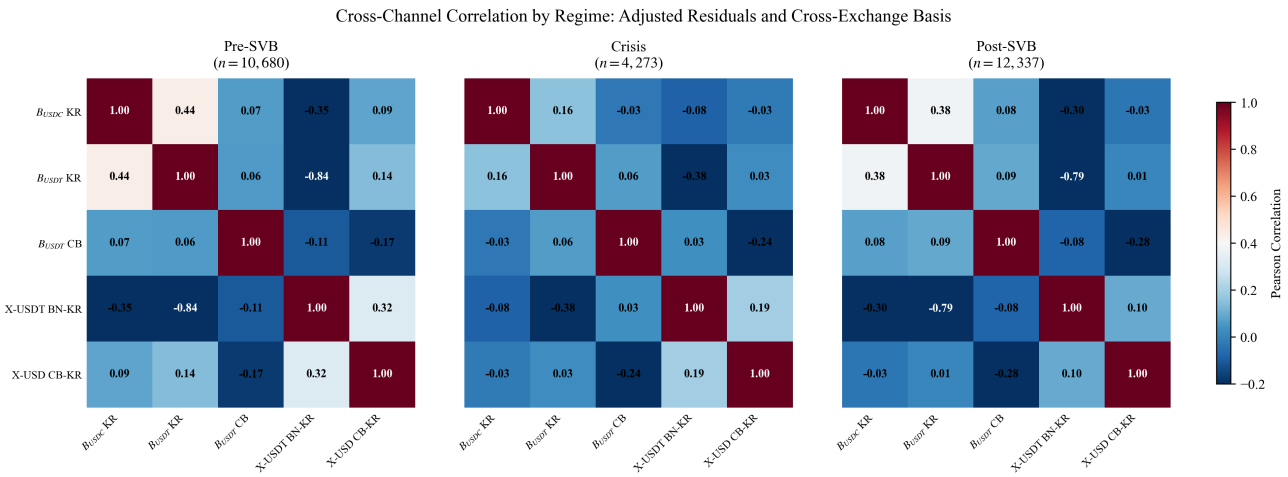

Figure 12: Cross-channel correlation by regime for adjusted residuals and cross-exchange basis. The B_{USDC}–B_{USDT} correlation collapses from 0.44 (pre-SVB) to 0.16 (crisis), consistent with divergent stablecoin dynamics rather than a common shock.

is significant ($q < 0.001$) but the reverse is not after FDR correction ($q = 0.13$), consistent with fiat-to-stablecoin price adjustment. All other channels (USDT intra-exchange, cross-exchange) show bidirectional significance, consistent with active two-way arbitrage in more liquid markets.

3.5 Arbitrage after fees

Having established that price discovery resides in the fiat-quoted channel, a natural question is whether the documented dislocations offer exploitable arbitrage opportunities. We evaluate this using explicit trade construction—3-leg triangular for intra-exchange stablecoin channels, 2-leg pre-funded buy/sell for cross-exchange—under two cost bounds:

$$\text{cost}_t^{\text{fee}} = n_{\text{legs}} \cdot f, \tag{6}$$

$$\text{cost}_t^{\text{fee+slip}} = n_{\text{legs}} \cdot f + \sum_{\ell=1}^{n_{\text{legs}}} \frac{1}{2}\,\text{RangeProxy}_{\ell,t}, \tag{7}$$

with $f = 5$ bps per taker leg.

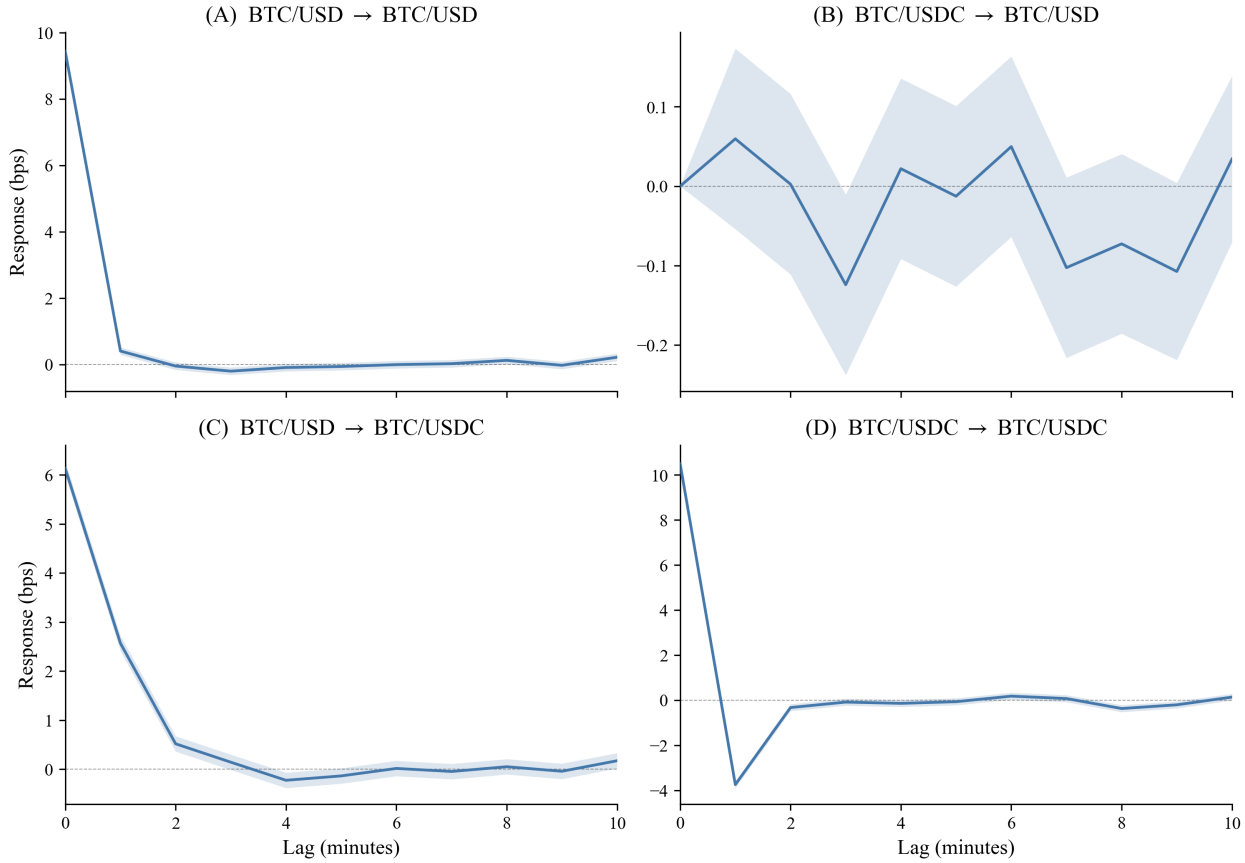

Figure 13: Orthogonalized impulse-response functions from bivariate VAR(10) on Kraken BTC/USD and BTC/USDC 1-minute returns (bps). (A) Own BTC/USD persistence. (B) BTC/USDC → BTC/USD: insignificant (CI includes zero). (C) BTC/USD → BTC/USDC: significant transmission (≈6 bps, decays in 3–4 min). (D) Own BTC/USDC response with partial reversal at lag 1. Shaded: 95% asymptotic CI.

Table 8: Arbitrage Profitability by Channel and Regime (5 bps/leg; 3-leg intra-exchange, 2-leg cross-exchange)

Channel	Regime	Cost Variant	%Profitable	AvgNetUncond (bps)
USDC/USD (Kraken)	Crisis	Fee-only	29.16	4.83
USDC/USD (Kraken)	Crisis	Fee+slip	6.72	0.64
USDC/USD (Kraken)	Post-SVB	Fee-only	9.38	0.56
USDC/USD (Kraken)	Post-SVB	Fee+slip	1.78	0.08
USDT/USD (Kraken)	Crisis	Fee-only	3.76	0.14
USDT/USD (Kraken)	Crisis	Fee+slip	0.37	0.01
Cross BTC/USD (CB–KR)	Crisis	Fee-only	11.32	0.72
Cross BTC/USD (CB–KR)	Crisis	Fee+slip	2.29	0.15
Cross BTC/USDT (BN–KR)	Crisis	Fee-only	11.85	0.49
Cross BTC/USDT (BN–KR)	Crisis	Fee+slip	1.81	0.05

Figure 14 shows the time series of net arbitrage profits under both cost specifications. Adding range-based slippage (Panel B) materially compresses profitability: USDC intra-exchange crisis profitability falls from 29.16% to 6.72% (net 4.83 to 0.64 bps); Coinbase–Kraken BTC/USD falls from 11.32% to 2.29%. Fee sensitivity: at $f=3$ bps (maker rebate), USDC crisis fee+slippage profitability rises to 14.0%; at $f=10$ bps (retail), it falls to 1.0%. The narrow post-cost opportunities are consistent with limited executable arbitrage under realistic frictions.

4 Regulatory Interpretation

The preceding analysis identifies a clear failure chain: reserve-access risk drives peg deviation, peg deviation drives dispersion, and limited depth constrains arbitrage correction. We now ask whether the GENIUS Act's

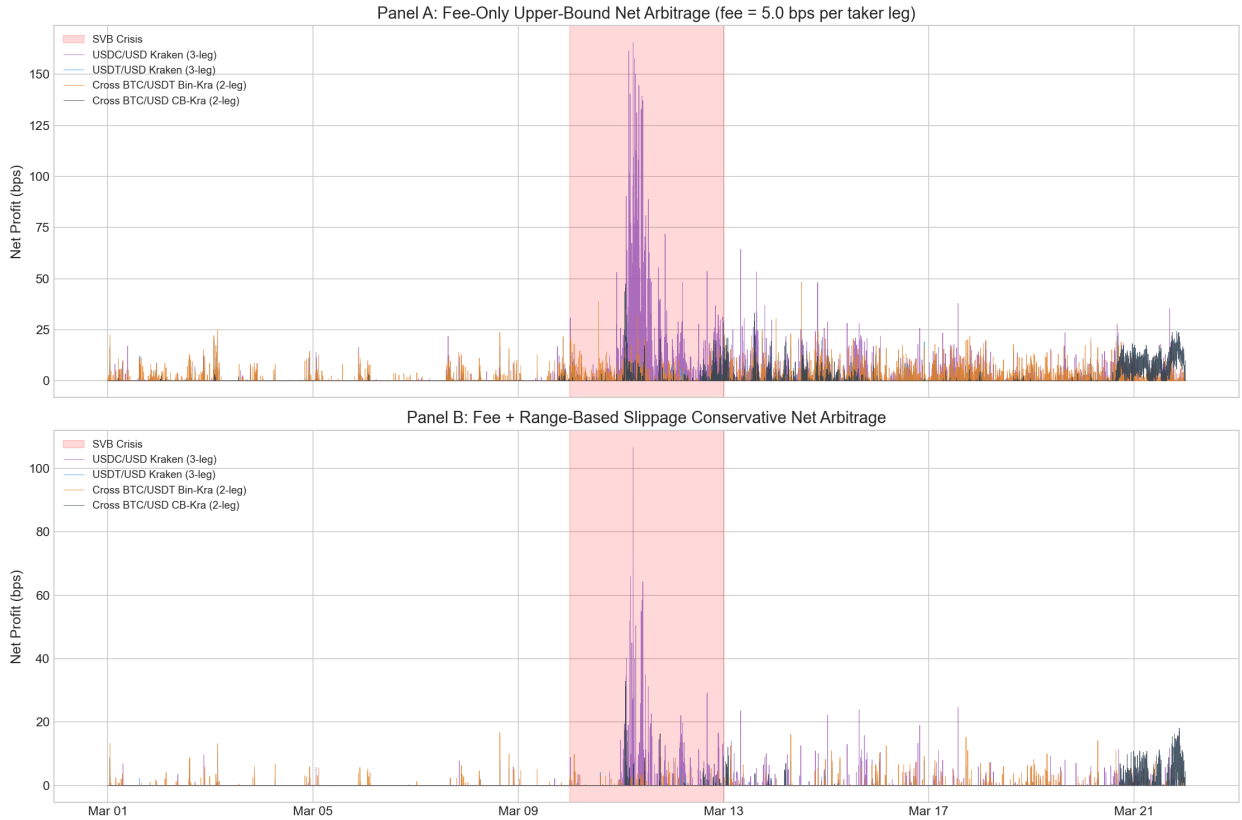

Figure 14: Arbitrage net profit (bps) over time. Panel A: fee-only (5 bps/leg); Panel B: fee + range-based slippage. The USDC intra-exchange channel (orange) spikes during crisis but collapses under slippage. Cross-exchange profits (dashed) are similarly attenuated.

provisions target this chain. The results point to banking-layer reserve-access risk as the root cause. Circle held \$3.3B (≈8%) of USDC reserves at SVB [3]; the $\sim 940\times$ half-life gap localizes persistent dislocation to the peg layer. Sustained recovery required emergency deposit guarantees on March 16 [5].

The GENIUS Act (July 2025) targets four dimensions of this chain; we map provisions to results as consistency checks [11]. (i) *Reserve composition*: Table 9 maps 25%–75% mitigation to implied D_t of $\approx$240–80 bps. (ii) *Timely redemption*: the 572-min peg half-life (vs. 1.06 pre-SVB) quantifies the five-day redemption freeze. (iii) *Bankruptcy super-priority*: targets P99 tail risk (81.4 vs. 13.6 bps). (iv) *Transparency*: traders learned of SVB exposure only from Circle's Friday-night disclosure, consistent with the USDC–USDT decorrelation (ρ=0.16).

Table 9: GENIUS Act scenario range (illustrative, not structural causal identification). The table maps assumed mitigation of reserve lock-up shock into implied crisis D_t means, anchored to observed pre- and crisis-period moments. Tail benchmark for context: USDC B_t crisis P99 = 81.4 bps versus pre-crisis P99 = 13.6 bps.

Scenario	Assumed mitigation of lock-up shock (%)	Implied crisis D_t mean (bps)	Reduction vs observed (bps)	Policy mapping
Observed baseline	0	320.0	0.0	SVB/No GENIUS baseline
Low mitigation	25	240.0	80.0	Reserve composition + redemption + transparency
Moderate mitigation	50	160.0	159.9	Reserve composition + redemption + transparency
High mitigation	75	80.1	239.9	Reserve composition + redemption + transparency
Full-mitigation lower bound	100	0.1	319.9	Reserve composition + redemption + transparency

Persistent cross-exchange deviations ($+2.18$ bps) and the $26\times$ depth gap suggest reserve-layer reforms alone may not eliminate spatial fragmentation.

5 Conclusion

Three findings emerge. First, SVB-triggered fragmentation is attributable to stablecoin peg risk: D_t reaches $+320$ bps but B_t stays near $+5$ bps with sub-minute mean reversion; the $\sim 940\times$ half-life gap localizes

persistent dislocation to reserve-access risk. Second, the crisis is USDC-specific: excess kurtosis 12.9 ($6\times$ tail expansion), USDT safe-haven premium ($+58$ bps), and cross-stablecoin decorrelation ($0.44 \rightarrow 0.16$). Third, post-friction arbitrage narrows sharply ($29\% \rightarrow 7\%$), with the $26\times$ depth gap limiting executable scale.

The GENIUS Act's provisions target precisely this failure chain. Because dislocation resides in the peg layer, reforms reducing reserve-access risk should compress D_t more effectively than measures targeting cross-market efficiency. However, persistent cross-exchange deviations and depth asymmetry suggest additional market-structure measures may also be needed. A natural next step is a post-GENIUS replication study during the next stress event.

Interpretation boundaries. This evidence is window-specific (March 2023) and observational; policy passages are consistency mappings, not causal identification. Robustness: (i) HAC intervals preserve directional signs; (ii) no-FF filters preserve rank-order conclusions; (iii) cointegration is rank-stable for BTC/USD–USDT but not USDC; (iv) Chow tests confirm the regime split.

Reproducibility. All exhibits are deterministic outputs of `run_all.py`; artifact-label integrity is verified by `src/04_tex_integrity_check.py`.

References

[1] Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.

[2] Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605.

[3] Circle (March 11, 2023). *Silicon Valley Bank and USDC*. Circle update.

[4] Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276.

[5] Federal Reserve, FDIC, and U.S. Treasury (March 12, 2023). *Joint statement by the Department of the Treasury, Federal Reserve, and FDIC*.

[6] Granger, C. W. J. (1969). Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3), 424–438.

[7] Hasbrouck, J. (1995). One security, many markets: Determining the contributions to price discovery. *Journal of Finance*, 50(4), 1175–1199.

[8] Johansen, S. (1988). Statistical analysis of cointegration vectors. *Journal of Economic Dynamics and Control*, 12(2–3), 231–254.

[9] Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

[10] Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance*, 39(4), 1127–1139.

[11] U.S. Congress (2025). *Public Law 119-27: Guiding and Establishing National Innovation for U.S. Stablecoins Act (GENIUS Act)*.